R116: Full18 Locked Results

Single-seed descriptive evidence. Historical models are not newly matched controls.

Model	Split	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
R116/A_static	VAL	70.897531	11.417557	11.427978	11.371692	-0.186796	0.400880	0.410795
R116/B_subset	VAL	70.933333	11.417765	11.427978	11.371692	-0.186384	0.401308	0.412386
R116/A_static	TEST	71.134568	11.374935	11.385299	11.329932	-0.186968	0.396260	0.404762
R116/B_subset	TEST	71.255556	11.374748	11.385299	11.329932	-0.190016	0.393213	0.402782
R94B	TEST	71.577778	11.372234	11.385299	11.329932	-0.211169	0.371939	0.379115

R116 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

R116: Corrections, Harms And Cost Gains

W=cost win rate; G=mean relative cost improvement. dRegret=current-baseline; lower is better.

Comparison	Split	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
B_subset vs A_static	VAL	779	750	371	459	0.035802	0.001591	1.4198	-0.002658
B_subset vs A_static	TEST	796	698	388	439	0.120988	-0.001980	1.4617	-0.001355
A_static vs R94B	TEST	1542	1901	752	785	-0.443210	0.025646	2.8309	-0.029081
B_subset vs R94B	TEST	1535	1796	678	782	-0.322222	0.023666	2.7309	-0.026918

A_static: VAL Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.444444	8.026467	8.045354	7.990533	-0.234757	0.449704	0.437783
CVRP	45.577778	18.163989	18.269377	18.064087	-0.576857	0.553042	0.721620
ATSP	77.244444	1.638586	1.676905	1.632471	-2.285085	0.374609	0.371184
OVRP	40.155556	7.463435	7.469579	7.415777	-0.082255	0.642663	0.658930
VRPB	81.022222	9.356323	9.354876	9.318750	0.015468	0.403203	0.408537
VRPL	38.466667	12.431605	12.437959	12.383444	-0.051091	0.388913	0.394818
VRPTW	79.955556	18.818827	18.819431	18.766139	-0.003211	0.280761	0.296633
OVRPTW	83.866667	10.142980	10.143730	10.116859	-0.007391	0.258191	0.254599
OVRPB	88.511111	6.357217	6.357254	6.341078	-0.000592	0.254511	0.264394
OVRPL	39.733333	7.439423	7.445709	7.390416	-0.084426	0.663123	0.684160
VRPBL	78.844444	9.369325	9.371523	9.328366	-0.023446	0.439081	0.443038
VRPBTW	78.511111	19.950453	19.950430	19.851084	0.000116	0.500576	0.482367
VRPLTW	78.800000	18.764975	18.765871	18.707843	-0.004777	0.305391	0.320967
OVRPBL	89.288889	6.369226	6.370093	6.352824	-0.013616	0.258183	0.267387
OVRPBTW	83.533333	10.660414	10.661098	10.623245	-0.006417	0.349886	0.322354
OVRPLTW	83.377778	10.158219	10.157423	10.129818	0.007836	0.280367	0.286573
VRPBLTW	79.555556	19.794509	19.797044	19.706420	-0.012804	0.447006	0.434132
OVRPBLTW	83.266667	10.610059	10.609955	10.571302	0.000975	0.366623	0.344827
ALL	70.897531	11.417557	11.427978	11.371692	-0.186796	0.400880	0.410795

R116 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

A_static: TEST Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.666667	7.972686	7.989621	7.937593	-0.211965	0.442104	0.432992
CVRP	46.600000	18.015332	18.124983	17.918509	-0.604972	0.540355	0.701358
ATSP	76.955556	1.635325	1.674207	1.629441	-2.322434	0.361071	0.360569
OVRP	40.088889	7.447169	7.455750	7.400487	-0.115092	0.630787	0.643609
VRPB	80.400000	9.357514	9.351810	9.315298	0.060995	0.453190	0.443758
VRPL	37.466667	12.349755	12.356138	12.300958	-0.051662	0.396687	0.404627
VRPTW	80.955556	18.690330	18.690545	18.637408	-0.001150	0.283957	0.294504
OVRPTW	83.933333	10.093026	10.093063	10.068135	-0.000365	0.247224	0.237943
OVRPB	88.066667	6.372354	6.370826	6.352225	0.023990	0.316874	0.334126
OVRPL	41.222222	7.464284	7.472542	7.418033	-0.110512	0.623486	0.635202
VRPBL	79.533333	9.395565	9.396379	9.355967	-0.008664	0.423237	0.429393
VRPBTW	78.777778	19.665869	19.665725	19.577666	0.000733	0.450531	0.447834
VRPLTW	79.266667	18.802889	18.803136	18.745143	-0.001316	0.308058	0.323045
OVRPBL	89.222222	6.375906	6.375741	6.359310	0.002580	0.260962	0.272955
OVRPBTW	84.911111	10.658817	10.659591	10.624748	-0.007268	0.320650	0.291325
OVRPLTW	82.911111	10.085981	10.086112	10.059001	-0.001291	0.268225	0.268565
VRPBLTW	79.777778	19.704089	19.707068	19.613284	-0.015116	0.462978	0.444833
OVRPBLTW	83.666667	10.661946	10.662149	10.625574	-0.001906	0.342304	0.319077
ALL	71.134568	11.374935	11.385299	11.329932	-0.186968	0.396260	0.404762

R116 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

B_subset: VAL Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	46.355556	8.026220	8.045354	7.990533	-0.237827	0.446613	0.435546
CVRP	45.600000	18.170163	18.269377	18.064087	-0.543061	0.587221	0.771454
ATSP	77.155556	1.638620	1.676905	1.632471	-2.283029	0.376721	0.372894
OVRP	40.777778	7.462776	7.469579	7.415777	-0.091084	0.633770	0.651176
VRPB	81.044444	9.353902	9.354876	9.318750	-0.010415	0.377220	0.384135
VRPL	38.222222	12.431232	12.437959	12.383444	-0.054090	0.385901	0.392157
VRPTW	79.933333	18.819230	18.819431	18.766139	-0.001068	0.282910	0.298478
OVRPTW	83.800000	10.143589	10.143730	10.116859	-0.001389	0.264209	0.261592
OVRPB	88.511111	6.357217	6.357254	6.341078	-0.000592	0.254511	0.264394
OVRPL	40.311111	7.439871	7.445709	7.390416	-0.078418	0.669176	0.690936
VRPBL	78.800000	9.369426	9.371523	9.328366	-0.022374	0.440158	0.444384
VRPBTW	78.488889	19.949048	19.950430	19.851084	-0.006926	0.493498	0.472441
VRPLTW	78.755556	18.765392	18.765871	18.707843	-0.002550	0.307625	0.323242
OVRPBL	89.333333	6.369265	6.370093	6.352824	-0.013005	0.258796	0.268312
OVRPBTW	83.511111	10.660998	10.661098	10.623245	-0.000937	0.355386	0.329524
OVRPLTW	83.466667	10.157632	10.157423	10.129818	0.002064	0.274580	0.279626
VRPBLTW	79.466667	19.795488	19.797044	19.706420	-0.007861	0.451973	0.442044
OVRPBLTW	83.266667	10.609705	10.609955	10.571302	-0.002360	0.363276	0.340614
ALL	70.933333	11.417765	11.427978	11.371692	-0.186384	0.401308	0.412386

R116 | native/FP64 winners | TRAIN SBS | equal-task ALL | same locked weights; no TEST/LIB reselection

B_subset: TEST Per-Problem Metrics

ALL equal-problem; ratios of means differ from actual per-instance regret

Problem	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
TSP	47.088889	7.972264	7.989621	7.937593	-0.217250	0.436785	0.428184
CVRP	47.133333	18.020067	18.124983	17.918509	-0.578848	0.566780	0.749645
ATSP	76.888889	1.635342	1.674207	1.629441	-2.321413	0.362120	0.361030
OVRP	40.400000	7.447238	7.455750	7.400487	-0.114166	0.631720	0.645056
VRPB	80.555556	9.353757	9.351810	9.315298	0.020815	0.412852	0.414258
VRPL	37.822222	12.349507	12.356138	12.300958	-0.053664	0.394675	0.402292
VRPTW	81.022222	18.689447	18.690545	18.637408	-0.005877	0.279216	0.290963
OVRPTW	83.955556	10.092872	10.093063	10.068135	-0.001890	0.245695	0.236098
OVRPB	88.133333	6.370873	6.370826	6.352225	0.000742	0.293558	0.299695
OVRPL	41.555556	7.464073	7.472542	7.418033	-0.113332	0.620645	0.632076
VRPBL	79.511111	9.396178	9.396379	9.355967	-0.002135	0.429794	0.434066
VRPBTW	78.844444	19.663418	19.665725	19.577666	-0.011730	0.438012	0.436068
VRPLTW	79.333333	18.802118	18.803136	18.745143	-0.005414	0.303947	0.317801
OVRPBL	89.266667	6.375635	6.375741	6.359310	-0.001674	0.256697	0.267982
OVRPBTW	84.822222	10.659348	10.659591	10.624748	-0.002279	0.325655	0.297624
OVRPLTW	82.933333	10.085687	10.086112	10.059001	-0.004210	0.265298	0.265684
VRPBLTW	79.755556	19.705496	19.707068	19.613284	-0.007977	0.470150	0.450416
OVRPBLTW	83.577778	10.662150	10.662149	10.625574	0.000012	0.344229	0.321130
ALL	71.255556	11.374748	11.385299	11.329932	-0.190016	0.393213	0.402782

B_subset vs A_static: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	53	34	27	32	0.422222	-0.004808	1.7778	0.004274
CVRP	210	186	146	174	0.533333	0.048287	7.9111	-0.049871
ATSP	26	29	1	1	-0.066667	0.000462	0.6000	-0.001037
OVRP	151	137	65	70	0.311111	0.001447	4.8000	-0.002273
VRPB	15	8	3	1	0.155556	-0.029500	0.4000	0.001476
VRPL	134	118	80	81	0.355556	-0.002335	4.7556	0.001914
VRPTW	6	3	0	0	0.066667	-0.003541	0.1333	0.003127
OVRPTW	2	1	0	0	0.022222	-0.001845	0.0444	0.001596
OVRPB	3	0	0	0	0.066667	-0.034430	0.0667	0.014847
OVRPL	153	138	66	78	0.333333	-0.003126	4.8667	0.002020
VRPBL	8	9	0	2	-0.022222	0.004673	0.1778	-0.005294
VRPBTW	12	9	0	0	0.066667	-0.011767	0.2667	0.007538
VRPLTW	9	6	0	0	0.066667	-0.005244	0.2000	0.004545
OVRPBL	2	0	0	0	0.044444	-0.004973	0.0444	0.004421
OVRPBTW	0	4	0	0	-0.088889	0.006299	0.0000	-0.006299
OVRPLTW	4	3	0	0	0.022222	-0.002881	0.0889	0.002469
VRPBLTW	7	8	0	0	-0.022222	0.005582	0.1556	-0.005764
OVRPBLTW	1	5	0	0	-0.088889	0.002053	0.0222	-0.002082
ALL	796	698	388	439	0.120988	-0.001980	1.4617	-0.001355

A_static vs R94B: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	110	126	68	59	-0.355556	0.025107	3.9556	-0.025673
CVRP	284	368	230	257	-1.866667	0.032816	11.4222	-0.039950
ATSP	22	32	4	0	-0.222222	0.005847	0.5778	-0.006064
OVRP	341	358	152	159	-0.377778	0.023298	10.9556	-0.025450
VRPB	43	73	8	3	-0.666667	0.089375	1.1333	-0.092267
VRPL	229	238	148	154	-0.200000	0.004452	8.3778	-0.005044
VRPTW	25	43	0	3	-0.400000	0.036765	0.5556	-0.037139
OVRPTW	17	24	0	0	-0.155556	-0.010248	0.3778	-0.002732
OVRPB	15	27	2	5	-0.266667	0.076141	0.3778	-0.077637
OVRPL	289	315	127	133	-0.577778	0.015060	9.2222	-0.016887
VRPBL	47	87	8	5	-0.888889	0.061289	1.2222	-0.065494
VRPBTW	30	45	0	0	-0.333333	0.022397	0.6667	-0.023843
VRPLTW	22	32	0	1	-0.222222	0.011034	0.4889	-0.015430
OVRPBL	12	25	4	5	-0.288889	0.030524	0.3556	-0.032034
OVRPBTW	9	17	0	0	-0.177778	0.006852	0.2000	-0.009649
OVRPLTW	13	29	1	1	-0.355556	-0.008261	0.3111	-0.007121
VRPBLTW	25	47	0	0	-0.488889	0.025766	0.5556	-0.027287
OVRPBLTW	9	15	0	0	-0.133333	0.013422	0.2000	-0.013749
ALL	1542	1901	752	785	-0.443210	0.025646	2.8309	-0.029081

B_subset vs R94B: TEST Paired Changes

Both-wrong contributions are normalized by all problem rows; see comparison.csv for full precision.

Problem	Corrected	Harmed	Wrong cheaper	Wrong dearer	dTop1 pp	dRegret pp	W %	G %
TSP	117	114	64	62	0.066667	0.020299	4.0222	-0.020898
CVRP	271	331	191	252	-1.333333	0.081103	10.2667	-0.085075
ATSP	26	39	4	0	-0.288889	0.006309	0.6667	-0.006707
OVRP	328	331	132	147	-0.066667	0.024745	10.2222	-0.026661
VRPB	47	70	7	1	-0.511111	0.059876	1.2000	-0.064352
VRPL	233	226	143	157	0.155556	0.002117	8.3556	-0.002720
VRPTW	25	40	0	3	-0.333333	0.033224	0.5556	-0.033601
OVRPTW	17	23	0	0	-0.133333	-0.012092	0.3778	-0.000887
OVRPB	16	25	1	5	-0.200000	0.041711	0.3778	-0.043248
OVRPL	285	296	124	141	-0.244444	0.011934	9.0667	-0.013848
VRPBL	45	86	7	7	-0.911111	0.065962	1.1556	-0.068002
VRPBTW	35	47	0	0	-0.266667	0.010631	0.7778	-0.012347
VRPLTW	21	28	0	1	-0.155556	0.005790	0.4667	-0.010239
OVRPBL	13	24	4	5	-0.244444	0.025551	0.3778	-0.027505
OVRPBTW	8	20	0	0	-0.266667	0.013151	0.1778	-0.015679
OVRPLTW	13	28	1	1	-0.333333	-0.011142	0.3111	-0.004240
VRPBLTW	27	50	0	0	-0.511111	0.031348	0.6000	-0.032738
OVRPBLTW	8	18	0	0	-0.222222	0.015475	0.1778	-0.015778
ALL	1535	1796	678	782	-0.322222	0.023666	2.7309	-0.026918

R116: Same-Checkpoint Native LIB

TEST participated in earlier R&D; not a pristine blind test; no finish-stage reselection

Model	Split	Top1 %	mean cost	SBS	Oracle	vs SBS %	vs Oracle %	Regret %
R116/A_static	TSPLIB	22.448980	8.214213	8.240435	8.142167	-0.318210	0.884855	0.910295
R116/A_static	CVRPLIB	43.000000	67.623271	67.245364	66.945093	0.561981	1.013035	0.909205
R116/B_subset	TSPLIB	24.489796	8.211730	8.240435	8.142167	-0.348347	0.854354	0.901432
R116/B_subset	CVRPLIB	44.000000	67.633942	67.245364	66.945093	0.577851	1.028976	0.921515

R116: Provenance And Scientific Boundary

same seeded task/shuffle protocol; actual per-update sample IDs not independently archived

Item	Evidence
Logical batch	1024
Initial trainable SHA	a6f46342085126b7e6cb452b21ee66b0ee0de59d0f1eaf0fba8e2f96bd81e46f
A_static checkpoint	c8a1288efe99aab41f56d97a409fdb69aee46c987438dfc660cf849a95d74f1a
B_subset checkpoint	d3e19c297fe94b31080393ab28b7171581bed449afd16cad35250d225dacba26
Data version	selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018
R77	missing_or_incompatible: Incomplete or inconsistent full18 frozen model: /public/home/shiys/0selection2/code/V4/runs/R77_sample_size_batch1024/B_
R94B	complete:
Seed	2 only; no multi-seed significance claim
Mechanism	identical fresh tensors/shapes; static versus subset-conditioned exact elimination
Source of metrics	immutable saved predictions; no new forward passes or calibration